

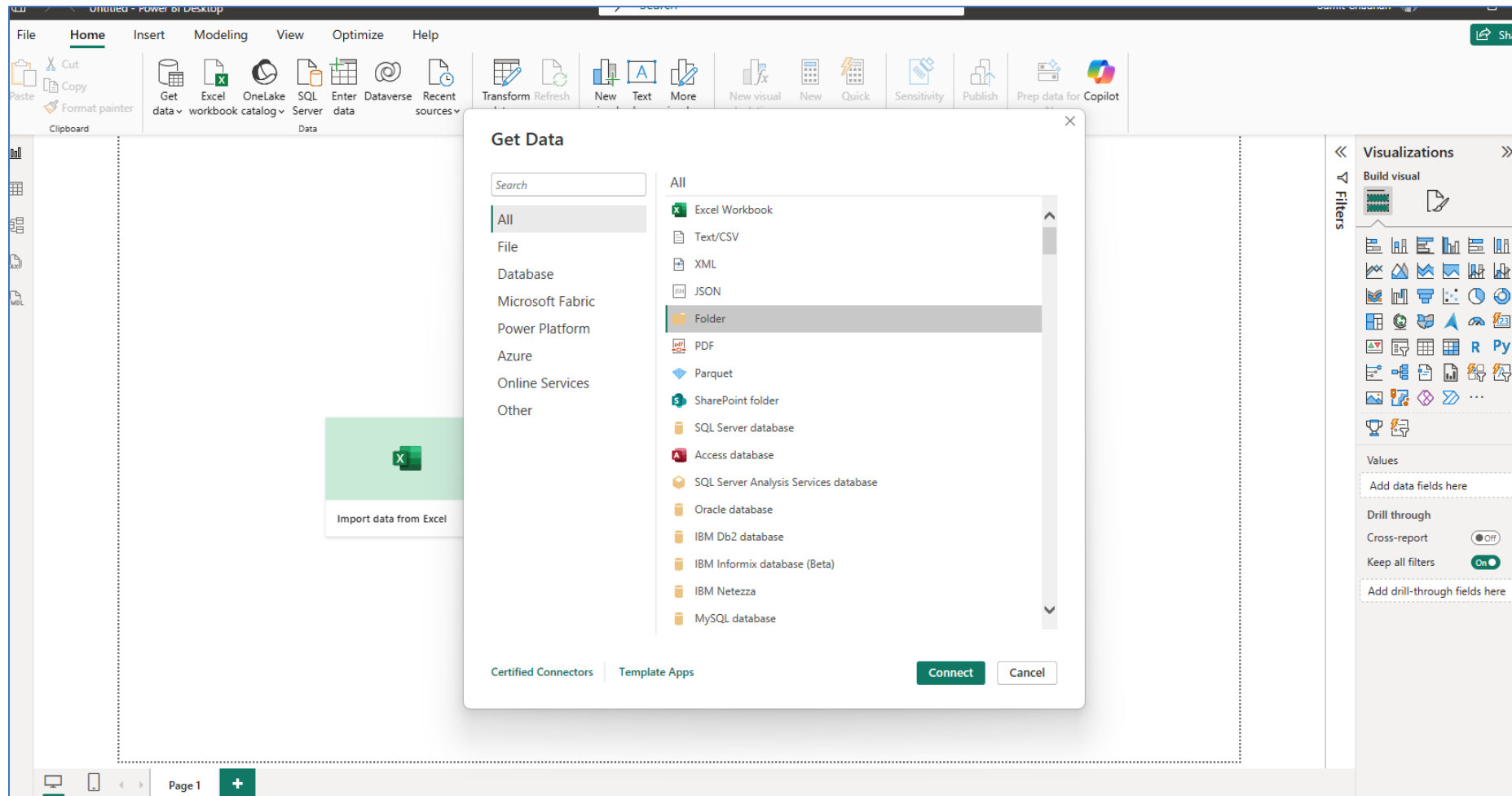
Power Query

Name: - Sumit Chauhan

Dept: - Data Engineering

Power Query Assignment

- Create queries to connect to all .csv files in the folder into power query



Removing unnecessary columns

The screenshot displays the Power BI Desktop interface with the Power Query Editor open. The formula bar at the top shows the M code: `= Table.RemoveColumns(Source,{"Extension", "Date accessed", "Date modified", "Date created", "Folder Path"})`. Below the formula bar, a table with 3 columns and 3 rows is visible. The columns are labeled 'Content', 'Name', and 'Attributes'. The rows represent different CSV files.

	Content	Name	Attributes
1	Binary	AdventureWorks Sales Data 2020.csv	Record
2	Binary	AdventureWorks Sales Data 2021.csv	Record
3	Binary	AdventureWorks Sales Data 2022.csv	Record

On the right side of the editor, the 'Query Settings' pane is open, showing the 'Properties' tab with the query name 'Sales Data' and the 'Applied Steps' list containing 'Source' and 'Removed Columns'.

At the bottom of the editor, a status bar indicates '3 COLUMNS, 3 ROWS' and 'Column profiling based on top 1000 rows'. The bottom of the Power BI Desktop window shows 'Page 1' and a zoom level of '96%'.

Keeping Sales_year column and merging sales data files

The screenshot displays the Microsoft Power Query Editor window. The main area shows a data table with the following columns: Sales_year, OrderDate, StockDate, OrderNumber, ProductKey, CustomerKey, and Territory. The data is organized into rows, with the first row being the header. The Sales_year column contains the year 2020 for all rows. The OrderDate, StockDate, OrderNumber, ProductKey, CustomerKey, and Territory columns contain various values representing sales data.

The ribbon at the top includes the following tabs: File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is currently selected, showing options like Merge Queries, Append Queries, and Combine Files. The Add Column tab is also visible, showing options like Data source settings, Manage Parameters, and Export query results.

The right-hand pane shows the Query Settings for the 'Sales Data' query. The Name field is set to 'Sales Data'. The APPLIED STEPS list includes the following steps: Source, Removed Columns, Split Column by Character Tra..., Split Column by Character Tra..., Removed Columns1, Changed Type, Renamed Columns, Filtered Hidden Files1, Invoke Custom Function1, Removed Columns2, and Expanded Transform File (2). The 'Expanded Transform File (2)' step is currently selected.

The bottom status bar indicates '9 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'. The bottom right corner shows 'PREVIEW DOWNLOADED AT 11:27'.

	Sales_year	OrderDate	StockDate	OrderNumber	ProductKey	CustomerKey	Territory
1	2020	2020-01-01	2019-09-21	SO45080	332	14657	1
2	2020	2020-01-01	2019-12-05	SO45079	312	29255	4
3	2020	2020-01-01	2019-10-29	SO45082	350	11455	9
4	2020	2020-01-01	2019-11-16	SO45081	338	26782	6
5	2020	2020-01-02	2019-12-15	SO45083	312	14947	10
6	2020	2020-01-02	2019-10-12	SO45084	310	29143	4
7	2020	2020-01-02	2019-12-18	SO45086	314	18747	9
8	2020	2020-01-02	2019-10-09	SO45085	312	18746	9
9	2020	2020-01-03	2019-10-03	SO45093	312	18906	9
10	2020	2020-01-03	2019-09-29	SO45090	310	29170	4
11	2020	2020-01-03	2019-12-11	SO45088	345	11398	10
12	2020	2020-01-03	2019-10-24	SO45092	313	18899	9
13	2020	2020-01-03	2019-12-16	SO45089	351	25977	4
14	2020	2020-01-03	2019-10-26	SO45091	314	18909	9
15	2020	2020-01-03	2019-09-11	SO45087	350	11388	10
16	2020	2020-01-03	2019-09-11	SO45094	310	22785	6
17	2020	2020-01-04	2019-10-30	SO45096	312	12483	7
18	2020	2020-01-04	2019-10-30	SO45097	313	29151	4
19	2020	2020-01-04	2019-09-15	SO45098	310	29167	1

- Add a new column to extract all characters before the dash (""-"") in the Product SKU column, and name it “SKU Type”

The screenshot displays the Microsoft Power Query Editor interface. The main window shows a table with columns 'ProductSKU' and 'ProductColor'. The 'ProductSKU' column contains values like 'HL-U509-R', 'HL-U509', 'SO-B909-M', etc. The 'ProductColor' column contains values like 'Red', 'Black', 'White', 'Blue', 'Multi', etc.

A 'Custom Column' dialog box is open in the center. It has the following fields:

- New column name:** SKUType
- Custom column formula:** `= Text.BeforeDelimiter([ProductSKU], "-")`

Below the formula field, it says 'No syntax errors have been detected.' There are 'OK' and 'Cancel' buttons at the bottom right of the dialog.

The background table data is as follows:

	ProductSKU	ProductColor
1	HL-U509-R	Red
2	HL-U509	Black
3	SO-B909-M	White
4	SO-B909-L	White
5	HL-U509-B	Blue
6	CA-1098	Multi
7	LI-0192-S	Multi
8	LI-0192-M	Multi
9	LI-0192-L	Multi
10	LI-0192-X	Multi
11	FR-R92R-62	Red
12	FR-R92R-44	Red
13	FR-R92R-48	Red
14	FR-R92R-52	Red
15	FR-R92R-56	Red
16	FR-R38B-58	Black
17	FR-R38B-60	Black
18	FR-R38B-62	Black
19	FR-R38R-44	Red
20	FR-R38R-48	Red
21	FR-R38R-52	Red

lms_practical

FileHomeTransformAdd ColumnViewToolsHelp

Column From Custom Invoke Custom Examples Column Function

Conditional ColumnIndex ColumnDuplicate Column

FormatMerge ColumnsABC123ExtractParseFrom Text

StatisticsStandardScientificFrom Number

TrigonometryRoundingInformationFrom Date & Time

Queries [10]

Transform File from S...
Transform File from S...
Other Queries [2]
Sales Data
AdventureWorks Pr...

CategoryKey

ProductSKU

SKUType

ProductName

ModelName

ProductDescription

1	31	HL-U509-R	HL	Sport-100 Helmet, Red	Sport-100	Universal fit, well-vented, lightweight , snap-on vis
2	31	HL-U509	HL	Sport-100 Helmet, Black	Sport-100	Universal fit, well-vented, lightweight , snap-on vis
3	23	SO-B909-M	SO	Mountain Bike Socks, M	Mountain Bike Socks	Combination of natural and synthetic fibers stays d
4	23	SO-B909-L	SO	Mountain Bike Socks, L	Mountain Bike Socks	Combination of natural and synthetic fibers stays d
5	31	HL-U509-B	HL	Sport-100 Helmet, Blue	Sport-100	Universal fit, well-vented, lightweight , snap-on vis
6	19	CA-1098	CA	AWC Logo Cap	Cycling Cap	Traditional style with a flip-up brim; one-size fits al
7	21	LJ-0192-S	LJ	Long-Sleeve Logo Jersey, S	Long-Sleeve Logo Jersey	Unisex long-sleeve AWC logo microfiber cycling jer
8	21	LJ-0192-M	LJ	Long-Sleeve Logo Jersey, M	Long-Sleeve Logo Jersey	Unisex long-sleeve AWC logo microfiber cycling jer
9	21	LJ-0192-L	LJ	Long-Sleeve Logo Jersey, L	Long-Sleeve Logo Jersey	Unisex long-sleeve AWC logo microfiber cycling jer
10	21	LJ-0192-X	LJ	Long-Sleeve Logo Jersey, XL	Long-Sleeve Logo Jersey	Unisex long-sleeve AWC logo microfiber cycling jer
11	14	FR-R92R-62	FR	HL Road Frame - Red, 62	HL Road Frame	Our lightest and best quality aluminum frame mad
12	14	FR-R92R-44	FR	HL Road Frame - Red, 44	HL Road Frame	Our lightest and best quality aluminum frame mad
13	14	FR-R92R-48	FR	HL Road Frame - Red, 48	HL Road Frame	Our lightest and best quality aluminum frame mad
14	14	FR-R92R-52	FR	HL Road Frame - Red, 52	HL Road Frame	Our lightest and best quality aluminum frame mad
15	14	FR-R92R-56	FR	HL Road Frame - Red, 56	HL Road Frame	Our lightest and best quality aluminum frame mad
16	14	FR-R38B-58	FR	LL Road Frame - Black, 58	LL Road Frame	The LL Frame provides a safe comfortable ride, whi
17	14	FR-R38B-60	FR	LL Road Frame - Black, 60	LL Road Frame	The LL Frame provides a safe comfortable ride, whi
18	14	FR-R38B-62	FR	LL Road Frame - Black, 62	LL Road Frame	The LL Frame provides a safe comfortable ride, whi
19	14	FR-R38R-44	FR	LL Road Frame - Red, 44	LL Road Frame	The LL Frame provides a safe comfortable ride, whi
20	14	FR-R38R-48	FR	LL Road Frame - Red, 48	LL Road Frame	The LL Frame provides a safe comfortable ride, whi
21	14	FR-R38R-52	FR	LL Road Frame - Red, 52	LL Road Frame	The LL Frame provides a safe comfortable ride, whi

Query Settings

PROPERTIES

Name
AdventureWorks Product Lookup
All Properties

APPLIED STEPS

Source
Promoted Headers
Changed Type
Added Custom
Reordered Columns

12 COLUMNS, 293 ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:49

- Update the SKU Type calculation above to return all characters before second dash

The screenshot displays the Microsoft Power Query Editor interface. A 'Custom Column' dialog box is open in the center, allowing the user to add a new column. The dialog contains the following information:

- New column name:** SKUType
- Custom column formula:** `= Text.BeforeDelimiter([ProductSKU], "-", 1)`
- Available columns:** A list of columns from the source data, including ProductKey, ProductSubcategoryKey, ProductSKU, ProductName, ModelName, ProductDescription, and ProductColor.

Below the formula field, a green checkmark indicates: "No syntax errors have been detected." The background shows a data table with columns 'CategoryKey', 'ProductSKU', and 'ProductColor'. The 'APPLIED STEPS' pane on the right shows the sequence of transformations: Source, Promoted Headers, Changed Type, and 'Added Custom' (the current step).

At the bottom of the window, the status bar indicates: "12 COLUMNS, 293 ROWS Column profiling based on top 1000 rows". The bottom right corner shows: "PREVIEW DOWNLOADED AT 11:49".

- Replace zeros (0) in the Product Style column with “NA”

The screenshot displays the Power Query Editor interface. A 'Replace Values' dialog box is open in the foreground, allowing the user to replace the value '0' with 'NA' in the 'ProductStyle' column. The background shows a data table with the following columns: ProductColor, ProductSize, ProductStyle, ProductCost, and ProductPrice. The 'ProductStyle' column contains values like 'U', 'L', 'M', and '0'. The 'ProductCost' and 'ProductPrice' columns contain numerical values.

ProductColor	ProductSize	ProductStyle	ProductCost	ProductPrice
Red	0	0	13.0863	
Black	0	0	12.0278	
White	M	U	3.3963	
White	L	U	3.3963	
Blue	0	0	12.0278	
Multi	0	U	5.7052	
Multi	0	U	21.7344	
Red	0	0	12.0278	
Red	0	0	12.0278	
Red	0	0	12.0278	
Red	0	0	12.0278	
Red	0	0	12.0278	
Black	0	0	2.0278	
Black	0	0	2.0278	
Black	0	0	2.0278	
Red	0	0	3.0278	
Red	0	0	3.0278	
Red	0	0	3.0278	

Replace Values

Replace one value with another in the selected columns.

Value To Find: 0

Replace With: NA

Advanced options

OK Cancel

Query Settings

PROPERTIES

Name: AdventureWorks Product Lookup

APPLIED STEPS

Source

Promoted Headers

Changed Type

Added Custom

Reordered Columns

12 COLUMNS, 293 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:54

a) Implement text transformations

- Duplicate the email address column and name it “Domain Name”

The screenshot displays the Power Query Editor window for a file named 'lms_practical'. The ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The main area shows a data table with the following columns: Date, MaritalStatus, Gender, EmailAddress, DomainName, AnnualIncome, and TotalCl. The 'DomainName' column is a duplicate of the 'EmailAddress' column. The 'APPLIED STEPS' pane on the right lists the following steps: Source, Promoted Headers, Changed Type, Duplicated Column, Renamed Columns, and Reordered Columns. The 'Reordered Columns' step is currently selected.

Date	MaritalStatus	Gender	EmailAddress	DomainName	AnnualIncome	TotalCl
08-04-1966	M	M	jon24@adventure-works.com	jon24@adventure-works.com	90000	
14-05-1965	S	M	eugene10@adventure-works.com	eugene10@adventure-works.com	60000	
12-08-1965	M	M	ruben35@adventure-works.com	ruben35@adventure-works.com	60000	
15-02-1968	S	F	christy12@adventure-works.com	christy12@adventure-works.com	70000	
08-08-1968	S	F	elizabeth5@adventure-works.com	elizabeth5@adventure-works.com	80000	
05-08-1965	S	M	julio1@adventure-works.com	julio1@adventure-works.com	70000	
09-05-1964	M	M	marco14@adventure-works.com	marco14@adventure-works.com	60000	
07-07-1964	S	F	rob4@adventure-works.com	rob4@adventure-works.com	60000	
01-04-1964	S	M	shannon38@adventure-works.com	shannon38@adventure-works.com	70000	
06-02-1964	S	F	jacquelyn20@adventure-works.com	jacquelyn20@adventure-works.com	70000	
04-11-1963	M	M	curtis9@adventure-works.com	curtis9@adventure-works.com	60000	
18-01-1968	M	F	lauren41@adventure-works.com	lauren41@adventure-works.com	100000	
06-08-1968	M	M	ian47@adventure-works.com	ian47@adventure-works.com	100000	
09-05-1968	S	F	sydney23@adventure-works.com	sydney23@adventure-works.com	100000	
27-02-1979	S	F	chloe23@adventure-works.com	chloe23@adventure-works.com	30000	
28-04-1979	M	M	wyatt32@adventure-works.com	wyatt32@adventure-works.com	30000	
26-06-1944	S	F	shannon1@adventure-works.com	shannon1@adventure-works.com	20000	
09-10-1944	S	M	clarence32@adventure-works.com	clarence32@adventure-works.com	30000	
07-03-1978	S	M	luke18@adventure-works.com	luke18@adventure-works.com	40000	
20-09-1978	S	M	jordan73@adventure-works.com	jordan73@adventure-works.com	40000	
03-09-1978	S	F	destiny7@adventure-works.com	destiny7@adventure-works.com	40000	

4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 13:10

- In the new column, remove all text/characters except for the domain name

The screenshot shows the Power Query Editor interface. The main data table has the following columns: BirthDate, MaritalStatus, Gender, EmailAddress, DomainName, AnnualIncome, and TotalCl. The DomainName column is currently empty. A dialog box titled "Text After Delimiter" is open, asking for a delimiter to extract data from the EmailAddress column. The delimiter "@" is entered in the input field. The dialog also has an "Advanced options" link and "OK" and "Cancel" buttons.

BirthDate	MaritalStatus	Gender	EmailAddress	DomainName	AnnualIncome	TotalCl
08-04-1966	M	M	jon24@adventure-works.com		90000	
14-05-1965	S	M	eugene10@adventure-works.com		60000	
12-08-1965	M	M	ruben35@adventure-works.com		60000	
15-02-1968	S	F	christy12@adventure-works.com		70000	
08-08-1968	S	F	elizabeth5@adventure-works.com		80000	
05-08-1965	S	M	julio1@adventure-works.com		70000	
09-05-1964	M	M	marco14@adventure-works.com		60000	
07-07-1964	S	F	rob4@adventure-works.com		60000	
01-04-1964	S	M				
06-02-1964	S	F				
04-11-1963	M	M				
18-01-1968	M	F				
06-08-1968	M	M				
09-05-1968	S	F				
27-02-1979	S	F				
28-04-1979	M	M				
26-06-1944	S	F				
09-10-1944	S	M				
07-03-1978	S	M				
20-09-1978	S	M				
03-09-1978	S	F				

14 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 13:11

lms_practical

FileHomeTransformAdd ColumnViewToolsHelp

Group By

Use First Row as Headers

Count Rows

Transpose

Reverse Rows

Count Rows

Data Type: Text

Detect Data Type

Rename

Replace Values

Fill

Pivot Column

Unpivot Columns

Move

Convert to List

Split Column

Format

Parse

Merge Columns

Extract

Parse

Statistics

Standard

Scientific

10²

Rounding

Information

Trigonometry

Rounding

Information

Date

Time

Duration

Run R script

Run Python script

Scripts

Queries [11]

Transform File from S...

Helper Queries [3]

Sample File

Parameter1 (Sampl...

Transform File

Transform Sample File

Transform File from S...

Helper Queries [3]

Sample File (2)

Parameter2 (Sampl...

Transform File (2)

Transform Sample Fi...

Other Queries [3]

Sales Data

AdventureWorks Pr...

AdventureWorks Cu...

fx

= Table.TransformColumns("#Reordered Columns", {{ "DomainName", each Text.AfterDelimiter(_, "@"), type text}})

	Site	MaritalStatus	Gender	EmailAddress	DomainName	AnnualIncome	TotalCl
1	08-04-1966	M	M	jon24@adventure-works.com	adventure-works.com	90000	
2	14-05-1965	S	M	eugene10@adventure-works.com	adventure-works.com	60000	
3	12-08-1965	M	M	ruben35@adventure-works.com	adventure-works.com	60000	
4	15-02-1968	S	F	christy12@adventure-works.com	adventure-works.com	70000	
5	08-08-1968	S	F	elizabeth5@adventure-works.com	adventure-works.com	80000	
6	05-08-1965	S	M	julio1@adventure-works.com	adventure-works.com	70000	
7	09-05-1964	M	M	marco14@adventure-works.com	adventure-works.com	60000	
8	07-07-1964	S	F	rob4@adventure-works.com	adventure-works.com	60000	
9	01-04-1964	S	M	shannon38@adventure-works.com	adventure-works.com	70000	
10	06-02-1964	S	F	jacquelyn20@adventure-works.com	adventure-works.com	70000	
11	04-11-1963	M	M	curtis9@adventure-works.com	adventure-works.com	60000	
12	18-01-1968	M	F	lauren41@adventure-works.com	adventure-works.com	100000	
13	06-08-1968	M	M	ian47@adventure-works.com	adventure-works.com	100000	
14	09-05-1968	S	F	sydney23@adventure-works.com	adventure-works.com	100000	
15	27-02-1979	S	F	chloe23@adventure-works.com	adventure-works.com	30000	
16	28-04-1979	M	M	wyatt32@adventure-works.com	adventure-works.com	30000	
17	26-06-1944	S	F	shannon1@adventure-works.com	adventure-works.com	20000	
18	09-10-1944	S	M	clarence32@adventure-works.com	adventure-works.com	30000	
19	07-03-1978	S	M	luke18@adventure-works.com	adventure-works.com	40000	
20	20-09-1978	S	M	jordan73@adventure-works.com	adventure-works.com	40000	
21	03-04-1978	S	F	destiny7@adventure-works.com	adventure-works.com	40000	

Query Settings

PROPERTIES

Name

AdventureWorks Customer Lookup

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Duplicated Column

Renamed Columns

Reordered Columns

Extracted Text After Delimiter

4 COLUMNS, 889 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 13:15

- Removing .com from domain name

The screenshot shows the Microsoft Power Query Editor interface. The main window displays a table with columns: **InitialStatus**, **Gender**, **EmailAddress**, **DomainName**, **AnnualIncome**, **TotalChildren**, and **Edu**. The **DomainName** column contains values like "adventure-works.com". A formula bar at the top shows the query formula: `= Table.TransformColumns("#Reordered Columns", {"DomainName", each Text.AfterDelimiter(_, "@", type text)})`. A dialog box titled "Text Before Delimiter" is open, prompting the user to "Enter the delimiter that marks the end of what you would like to extract." The "Delimiter" field contains a period ".". The "Advanced options" section is expanded. The "Query Settings" pane on the right shows the "APPLIED STEPS" list, which includes "Extracted Text After Delimiter".

Text Before Delimiter

Enter the delimiter that marks the end of what you would like to extract.

Delimiter

.

Advanced options

OK Cancel

Query Settings

PROPERTIES

Name

AdventureWorks Customer Lookup

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Duplicated Column
- Renamed Columns
- Reordered Columns
- Extracted Text After Delimiter

14 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 13:1

- Replacing – with ' ' (space)

The screenshot shows the Microsoft Power Query Editor interface. The main window displays a data table with columns: **InitialStatus**, **Gender**, **EmailAddress**, **DomainName**, **AnnualIncome**, **TotalChildren**, and **Edu**. The **DomainName** column contains values like "adventure-works". A formula bar at the top shows a M query: `= Table.TransformColumns("#Extracted Text After Delimiter", {"DomainName", each Text.BeforeDelimiter(_, "."), type text})`.

A "Replace Values" dialog box is open in the foreground. It has the title "Replace Values" and the instruction "Replace one value with another in the selected columns." It contains two input fields: "Value To Find" with the text "-" and "Replace With" which is empty. There is an "Advanced options" link and "OK" and "Cancel" buttons at the bottom.

On the right side, the "Query Settings" pane is visible, showing the "PROPERTIES" section with the name "AdventureWorks Customer Lookup" and the "APPLIED STEPS" list, which includes "Extracted Text Before Delimiter" (highlighted with an 'X').

The status bar at the bottom indicates "4 COLUMNS, 999+ ROWS" and "Column profiling based on top 1000 rows". The bottom right corner shows "PREVIEW DOWNLOADED AT 13:24".

- Apply formatting to capitalize each word

The screenshot displays the Microsoft Power Query Editor interface. The 'Format' dropdown menu is open, showing options like lowercase, UPPERCASE, Capitalize Each Word, Trim, Clean, Add Prefix, and Add Suffix. The 'Capitalize Each Word' option is selected. The background shows a data table with columns: EmailAddress, Gender, AnnualIncome, TotalChildren, and Education. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Replaced Value'.

EmailAddress	Gender	AnnualIncome	TotalChildren	Education
jon24@adventure-works.com	M	90000	2	Bach
eugene10@adventure-works.com	M	60000	3	Bach
ruben35@adventure-works.com	M	60000	3	Bach
christy12@adventure-works.com	F	70000	0	Bach
elizabeth5@adventure-works.com	F	80000	5	Bach
julio1@adventure-works.com	M	70000	0	Bach
marco14@adventure-works.com	M	60000	3	Bach
rob4@adventure-works.com	F	60000	4	Bach
shannon38@adventure-works.com	M	70000	0	Bach
jacquelyn20@adventure-works.com	F	70000	0	Bach
curtis9@adventure-works.com	M	60000	4	Bach
lauren41@adventure-works.com	F	100000	2	Bach
ian47@adventure-works.com	M	100000	3	Bach
sydney23@adventure-works.com	F	100000	0	Parti
chloe23@adventure-works.com	F	30000	0	Parti
wyatt32@adventure-works.com	M	30000	0	Parti
shannon1@adventure-works.com	F	20000	4	High
clarence32@adventure-works.com	M	30000	2	Parti
luke18@adventure-works.com	M	40000	0	High
jordan73@adventure-works.com	M	40000	0	High
destiny7@adventure-works.com	F	40000	0	Parti

14 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 13:26

- Final Domain Name column

The screenshot shows the Microsoft Power Query Editor interface. The main area displays a table with the following columns: InitialStatus, Gender, EmailAddress, DomainName, AnnualIncome, TotalChildren, and Education. The 'DomainName' column is highlighted, and the formula bar shows the transformation: `= Table.TransformColumns("#Replaced Value",{{"DomainName", Text.Proper, type text}})`.

The 'APPLIED STEPS' pane on the right shows the following steps:

- Source
- Promoted Headers
- Changed Type
- Duplicated Column
- Renamed Columns
- Reordered Columns
- Extracted Text After Delimiter
- Extracted Text Before Delimiter
- Replaced Value
- Capitalized Each Word**

The status bar at the bottom indicates "14 COLUMNS, 999+ ROWS" and "Column profiling based on top 1000 rows". The preview was downloaded at 13:28.

b) Numeric tools

- What is our average product cost?

The screenshot displays the Microsoft Excel application window. The title bar at the top shows the Excel icon, a save icon, and the file name "lms_practical". The ribbon is set to the "Formulas" tab, with the "Number Tools" group expanded, showing options like "Trigonometry", "Rounding", and "Information". The "Formulas" tab ribbon also includes "File", "Home", "Transform", "Add Column", "View", "Tools", and "Help".

The main workspace is divided into two panes. The left pane, titled "Queries [11]", shows a tree view of queries. The right pane displays the formula bar and the result of the formula. The formula bar contains the formula `= List.Average(#"Replaced Value"[ProductCost])`. The result of the formula, `413.66100921501641`, is shown in the cell below the formula bar.

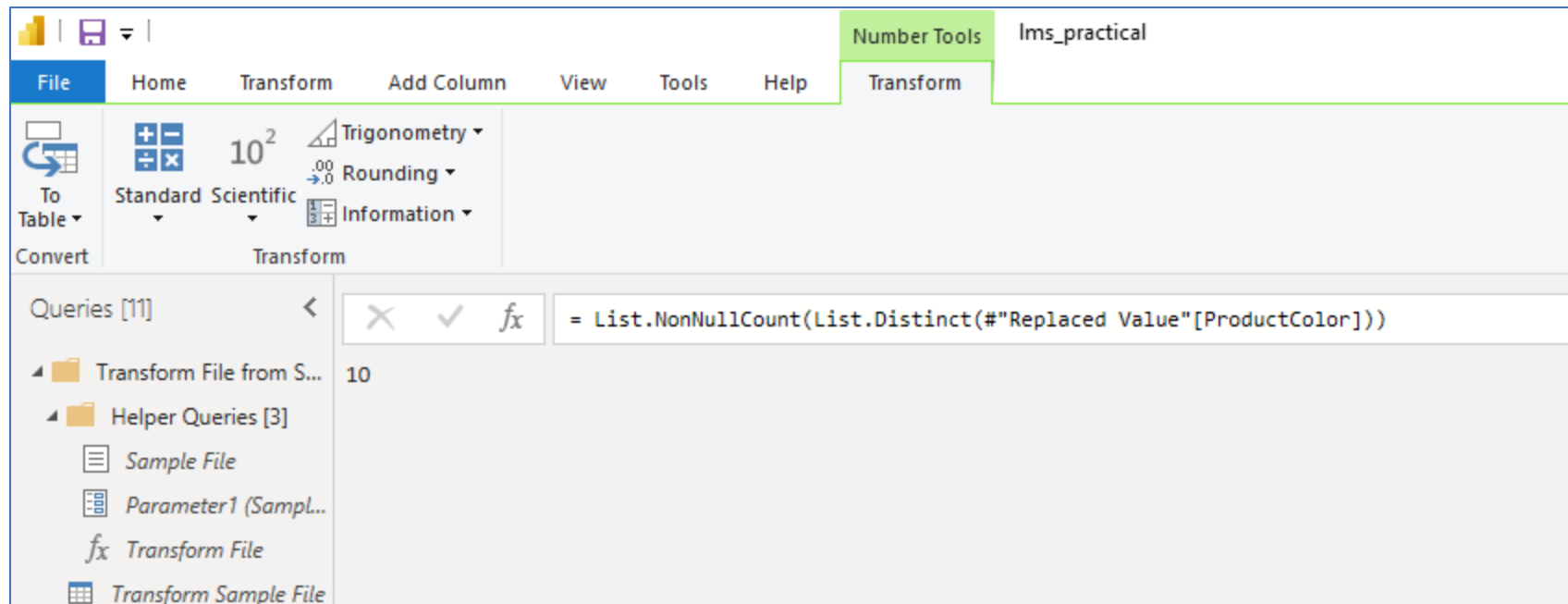
Queries [11]

- Transform File from S...
- Helper Queries [3]
 - Sample File
 - Parameter1 (Sampl...
 - Transform File
 - Transform Sample File
- Transform File from S...

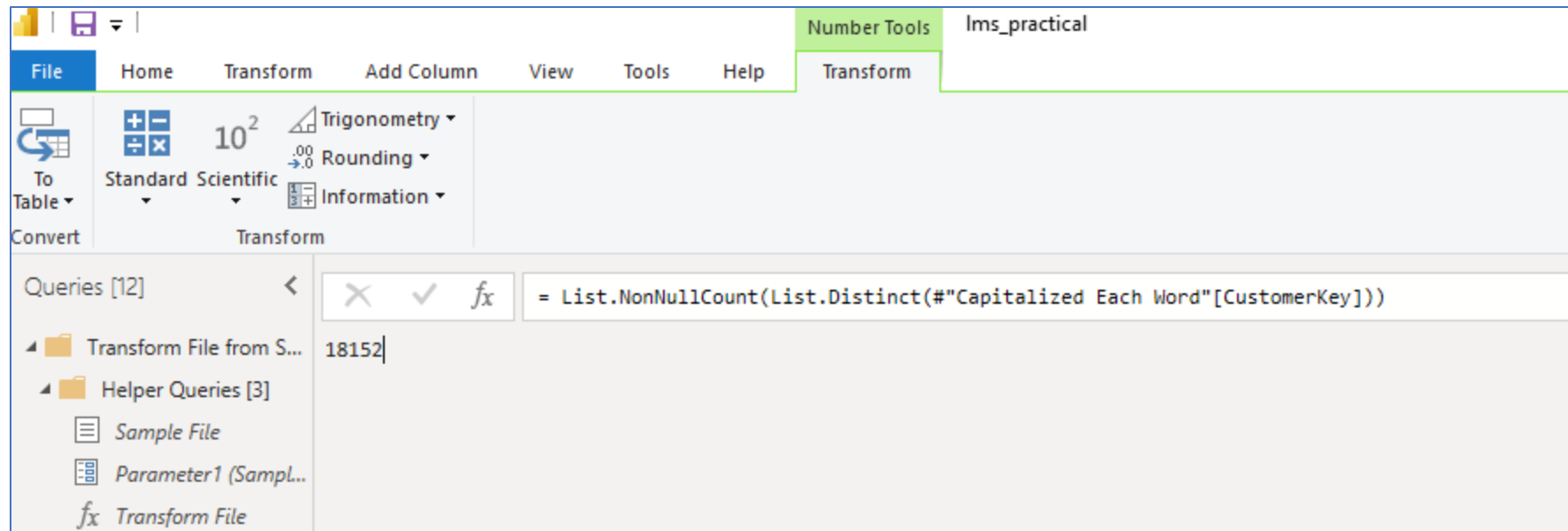
Formula Bar: `= List.Average(#"Replaced Value"[ProductCost])`

Result: 413.66100921501641

- How many colors do we sell our products in?



- How many distinct customers do we have?



- What is the maximum annual customer income?

The screenshot shows the Microsoft Excel interface with the 'Number Tools' tab selected. The ribbon includes 'File', 'Home', 'Transform', 'Add Column', 'View', 'Tools', and 'Help'. The 'Transform' sub-tab is active, showing options like 'To Table', 'Convert', 'Standard', 'Scientific', 'Trigonometry', 'Rounding', and 'Information'. The formula bar displays the formula `= List.Max(#"Capitalized Each Word"[AnnualIncome])`. The result of the formula, 170000, is shown in the cell below the formula bar. The left sidebar shows a list of queries, including 'Transform File from S...', 'Helper Queries [3]', 'Sample File', 'Parameter1 (Sampl...', 'Transform File', and 'Transform Sample File'.

Queries [11]	Result
Transform File from S...	
Helper Queries [3]	
Sample File	
Parameter1 (Sampl...	
Transform File	
Transform Sample File	

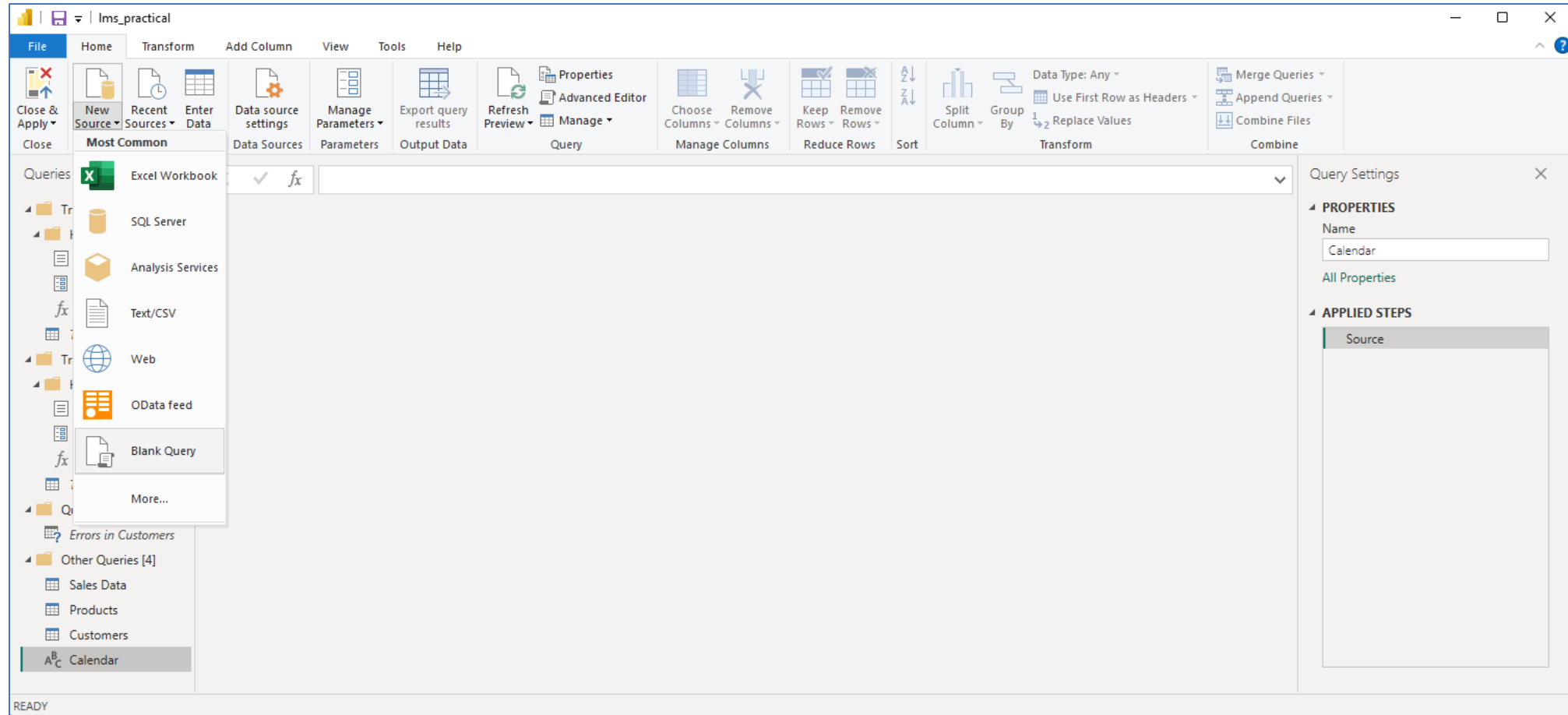
- Return the tables to their original state

The screenshot displays the Microsoft Power Query Editor window. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is active, showing various data manipulation options. The main area displays a data table with 13 columns and 999+ rows. The columns are labeled A1C Column1 through A1C Column7. The data includes customer information such as CustomerKey, Prefix, FirstName, LastName, BirthDate, MaritalStatus, and Gender. The right-hand pane shows the Query Settings for 'Customers', including the Name, Properties, and Applied Steps. The Applied Steps list includes Source, Promoted Headers, Changed Type, Duplicated Column, Renamed Columns, Reordered Columns, Extracted Text After Delimiter, Extracted Text Before Delimiter, Replaced Value, and Capitalized Each Word. The status bar at the bottom indicates '13 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

A1C Column1	A1C Column2	A1C Column3	A1C Column4	A1C Column5	A1C Column6	A1C Column7
1	CustomerKey	Prefix	FirstName	LastName	BirthDate	MaritalStatus
2	11000	MR.	JON	YANG	1966-04-08	M
3	11001	MR.	EUGENE	HUANG	1965-05-14	S
4	11002	MR.	RUBEN	TORRES	1965-08-12	M
5	11003	MS.	CHRISTY	ZHU	1968-02-15	S
6	11004	MRS.	ELIZABETH	JOHNSON	1968-08-08	S
7	11005	MR.	JULIO	RUIZ	1965-08-05	S
8	11007	MR.	MARCO	MEHTA	1964-05-09	M
9	11008	MRS.	ROBIN	VERHOFF	1964-07-07	S
10	11009	MR.	SHANNON	CARLSON	1964-04-01	S
11	11010	MS.	JACQUELYN	SUAREZ	1964-02-06	S
12	11011	MR.	CURTIS	LU	1963-11-04	M
13	11012	MRS.	LAUREN	WALKER	1968-01-18	M
14	11013	MR.	IAN	JENKINS	1968-08-06	M
15	11014	MRS.	SYDNEY	BENNETT	1968-05-09	S
16	11015	MS.	CHLOE	YOUNG	1979-02-27	S
17	11016	MR.	WYATT	HILL	1979-04-28	M
18	11017	MRS.	SHANNON	WANG	1944-06-26	S
19	11018	MR.	CLARENCE	RAI	1944-10-09	S
20	11019	MR.	LUKE	LAL	1978-03-07	S
21	11020	MR.	JORDAN	KING	1978-09-20	S
22	11021	MS.	DESTINY	WILSON	1978-09-03	S

c) Date Transformations

- Create a calendar table using M language (creating blank query for it)



- Advanced editor

